

WASP-12b

R-0246

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_241129 · created 2026-07-17T05:48:00+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460643.8426 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1374 +/- 0.0076
Transit depth (Rp/Rs)^2	1.89 +/- 0.21 [%]
Orbital Inclination (inc)	76.83 +/- 0.52
Airmass coefficient 1 (a1)	0.891 +/- 0.0065
Airmass coefficient 2 (a2)	0.114 +/- 0.039
Scatter in the residuals of the lightcurve fit is	0.73 %
Best Comparison Star	#1 - [63, 38]
Optimal Aperture	4.05
Optimal Annulus	13.62
Transit Duration (day)	0.0756 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1374 ± 0.0076 vs 0.1178 (deviation 2.02σ)
Duration fit vs published	0.0756 d vs 0.125 d (deviation 22.74σ)
Mid-time O–C	-3.9 min
Depth SNR	14.1
Residual scatter	0.73 %

Light curve

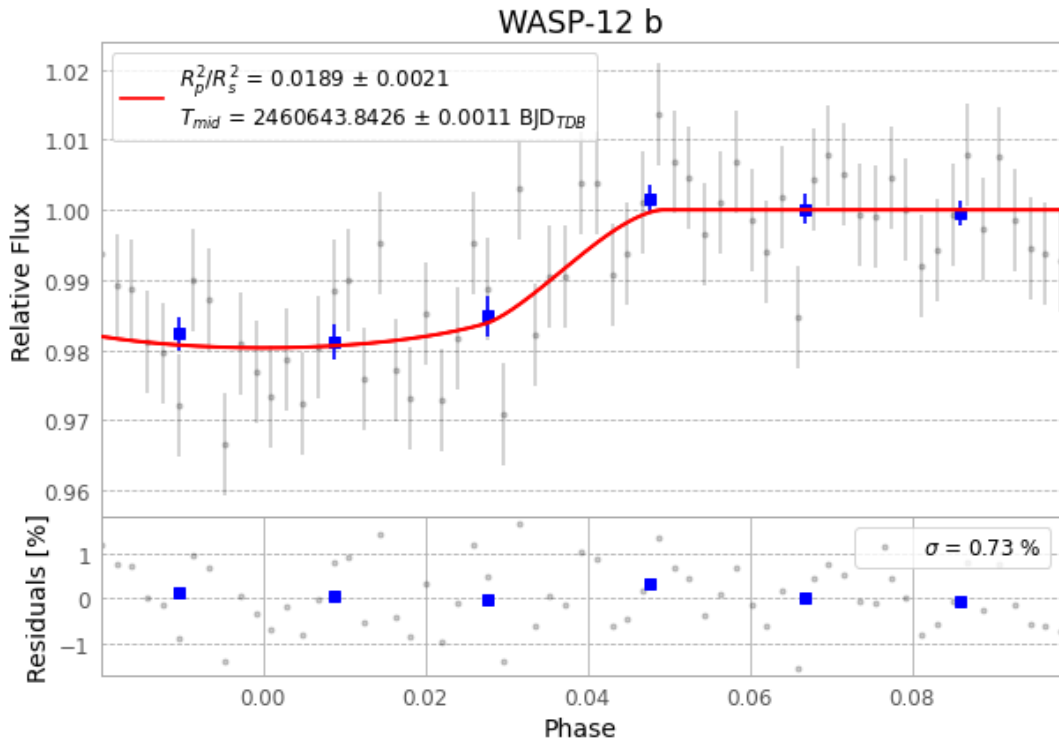


Figure 1 — FinalLightCurve_WASP-12 b_2024-11-29.png (SHA-256 dbd8b46259081bf7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [400, 241], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 497601bfa8384d0d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

64 FITS frames (42 MB), WASP-12241129073316.FITS → WASP-12241129104215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-241129.log (5dab8c02c7ff...), FinalLightCurve_WASP-12 b_2024-11-29.png (dbd8b4625908...), FinalLightCurve_WASP-12 b_2024-11-29.csv (d19ad43a2711...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 05:48Z.